STA/OPR 9750 Final Presentation

New York Minute:

Evaluating Emergency Response Across Boroughs

Team members (Group 3) :

Abdullah Alam, Noor Mahmud, Alba Marsela, Yashasvi Bhati, Moksha Shah

Our ‘Animating’ Question & Sub-Questions

Does your borough affect your longevity?

Which borough is reporting the most incidents?

Which borough has the least emergency response time taken?

Is there a correlation between the number of incidents and response time?

Data Sources

Emergency Response -
https://data.cityofnewyork.us/Public-Safety/Emergency-Response-Incidents/pasr-j7fb/about_data

Calls for Service / 911 -
https://data.cityofnewyork.us/Public-Safety/NYPD-Calls-for-Service-Year-to-Date-/n2zq-pubd/about_data

Emergency Response Incidents

The Process:

- Loaded data into R
- Performed statistical summary
- Dropped unnecessary records
- Goal: Borough with highest amount of incidents

Challenges:

- Null values
- Boroughs – typos
- Data type – different formats (string/numeric)

What's in this Dataset?

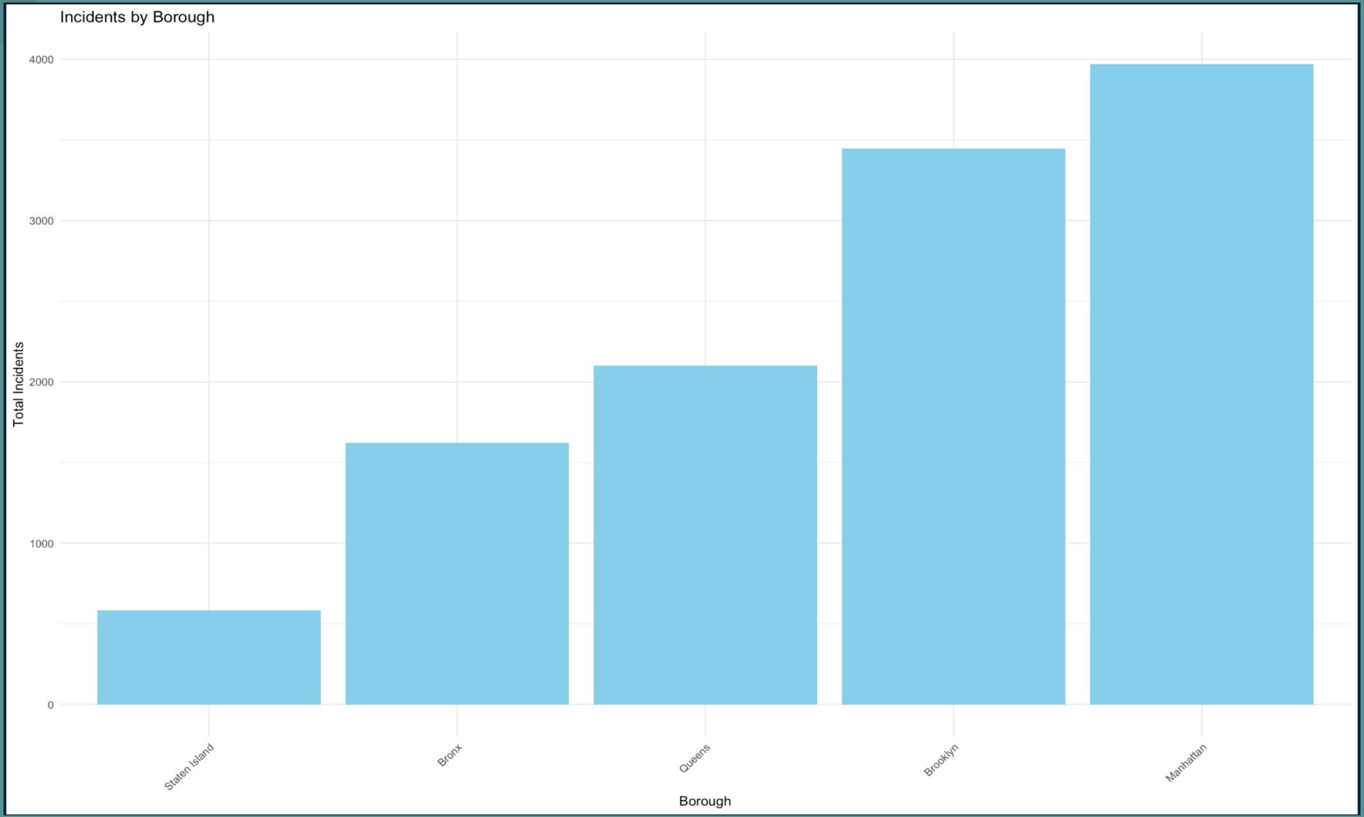
Rows	Columns
11.8K	7

Columns in this Dataset

Column Name
Incident Type
Location
Borough
Creation Date
Closed Date
Latitude
Longitude

Total number of Incidents Reported in each Borough

Manhattan has the highest number of incidents reported.



Critical Calls: Exploring the 911 Dataset

The Process:

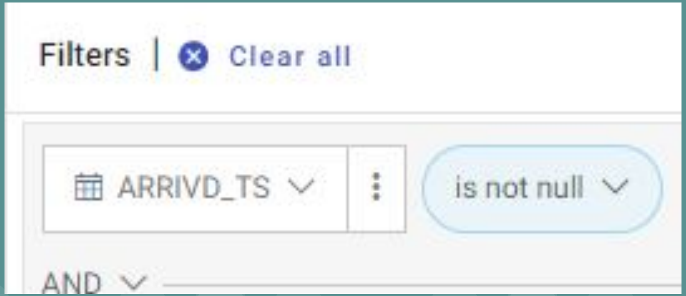
- We started off with 40.7 M rows!!
- We used NYC open data query and grouping
- We focused only on year 2023, CFS 911

The Criteria:

- Added to the system
- Dispatched to a unit
- NYPD arrived at the location
- The incident was closed

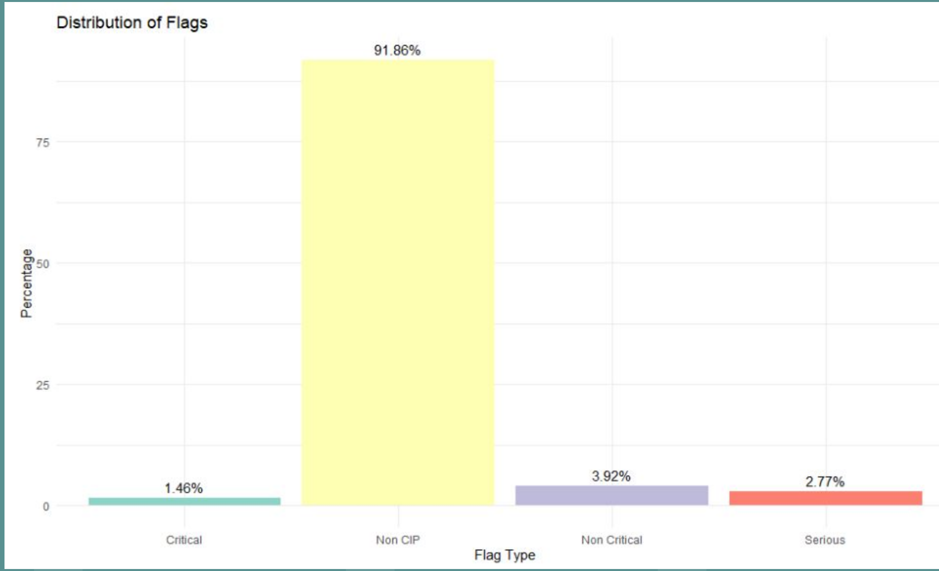
The Outcome:

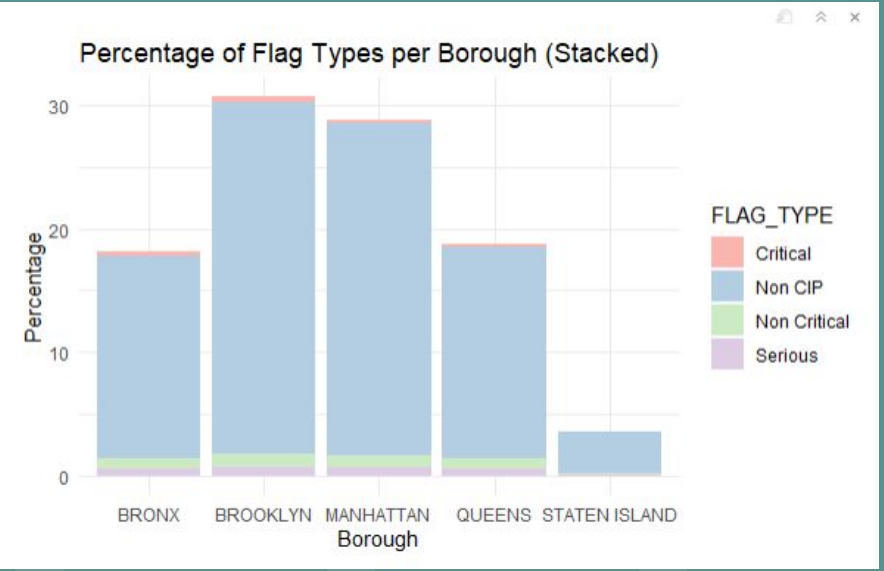
- The data ranged from 1 - 2 million rows per borough



Findings from Exploratory Data Analysis (EDA)

Breakdown of the flag type





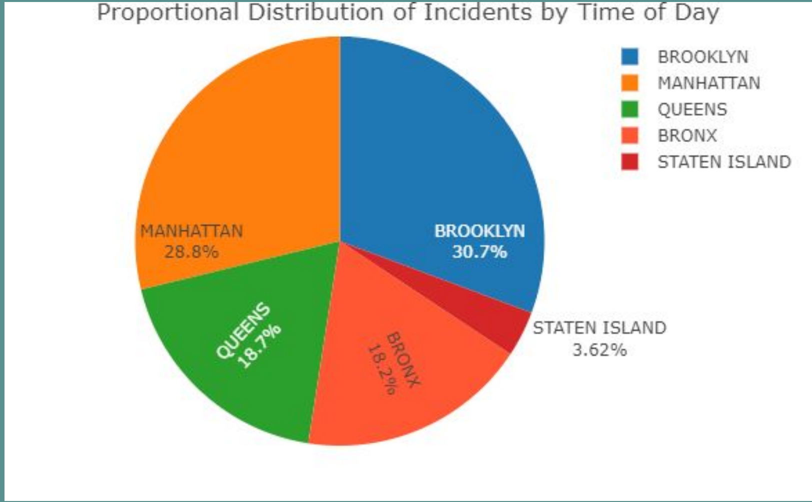
Findings from Exploratory Data Analysis (EDA)

Morning time incidents > Night time incidents

A tibble: 5 x 4

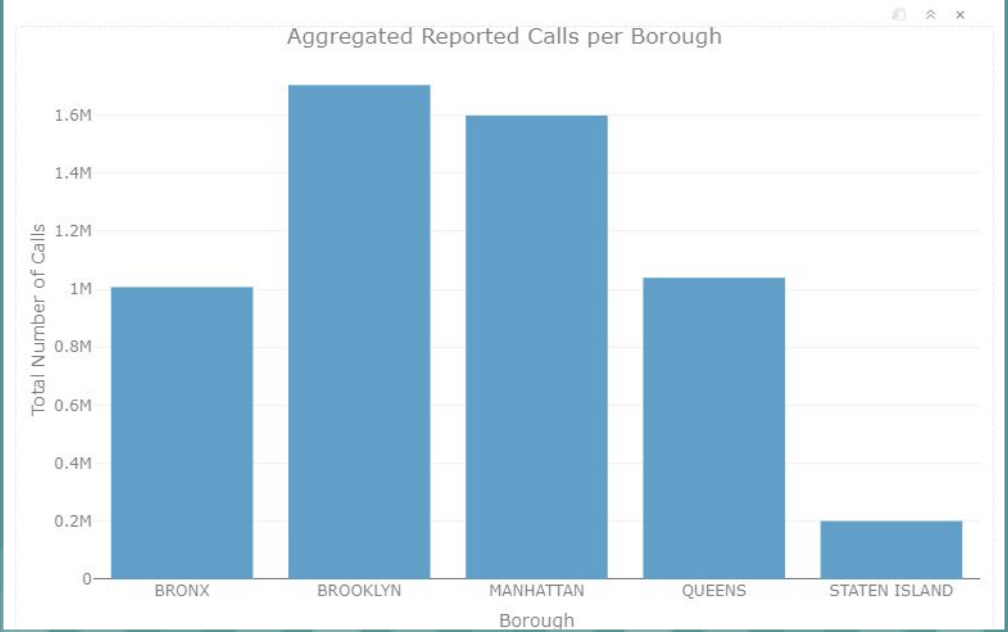
BORO_NM <chr>	Daytime <int>	Nighttime <int>	Daytime_to_Nighttime_Ratio <dbl>
BRONX	520177	486652	1.068889
BROOKLYN	913874	788603	1.158852
MANHATTAN	836435	761322	1.098661
QUEENS	546445	492462	1.109619
STATEN ISLAND	106210	94722	1.121281

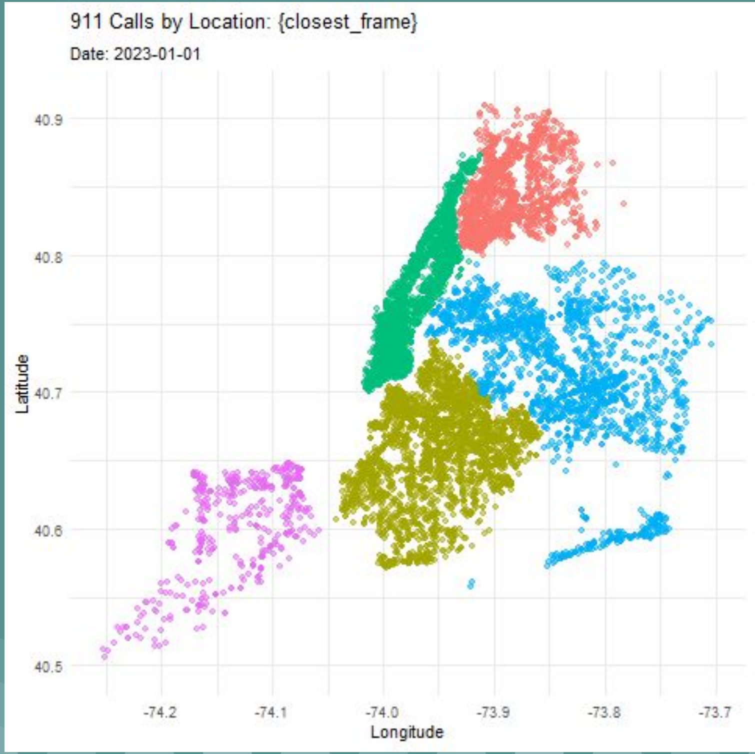
5 rows



Findings from Exploratory Data Analysis (EDA)

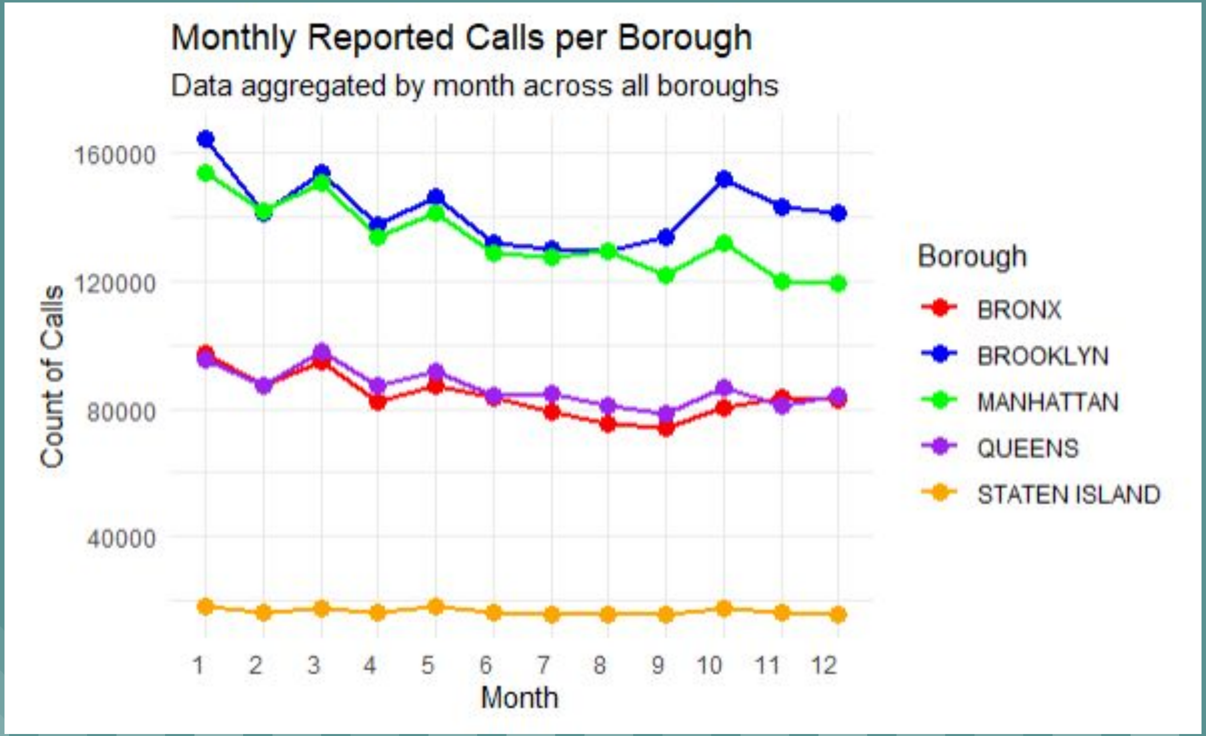
Brooklyn has the most 911 calls, as seen by the bar chart and scatter plot.



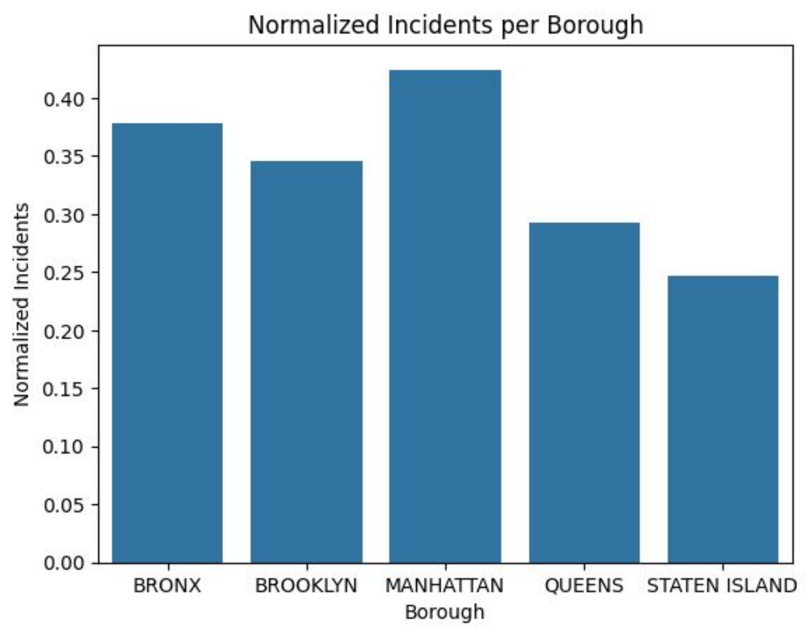


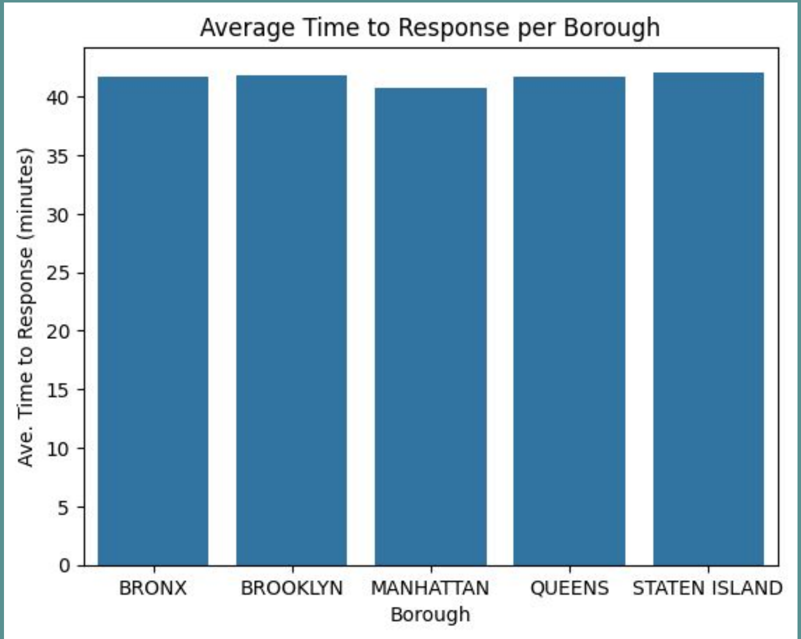
Findings from Exploratory Data Analysis (EDA)

Monthly fluctuations in 911 calls between Manhattan and Brooklyn

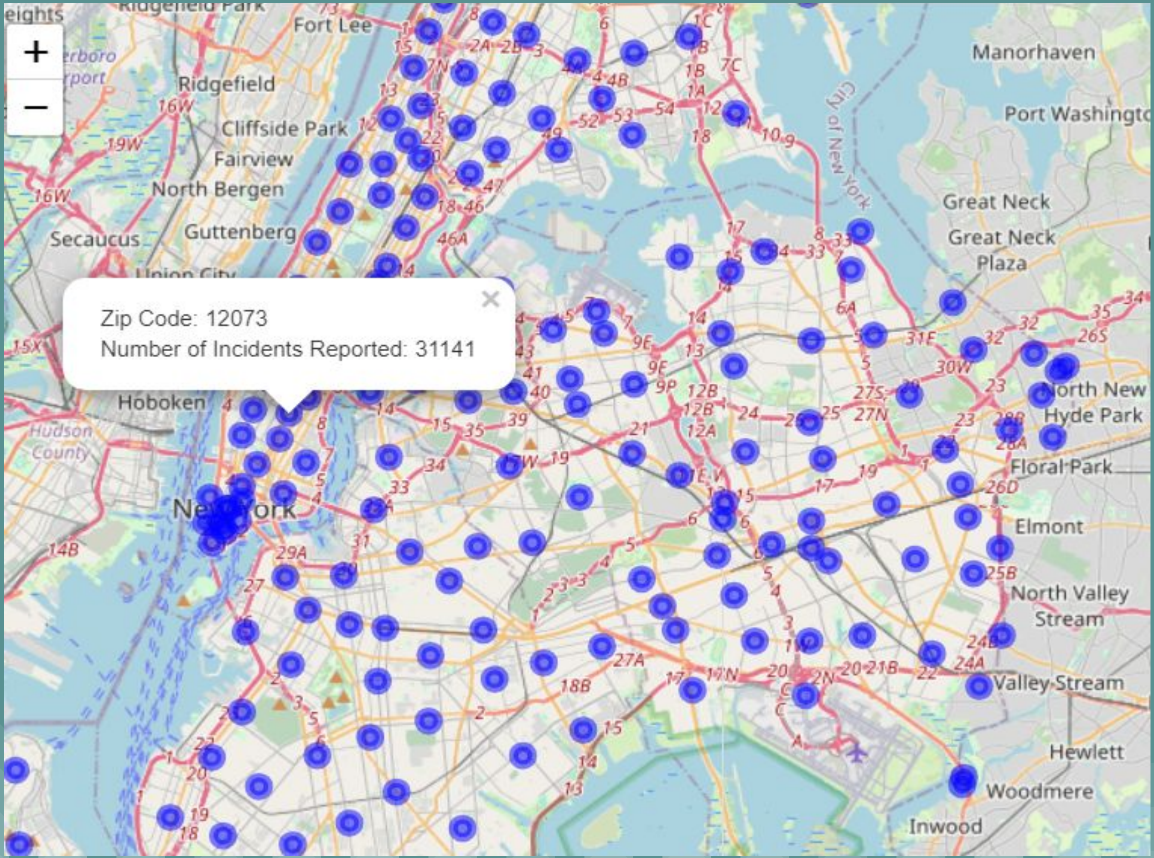


Normalization





Going Further With Our Data



Overall Findings

Manhattan has the highest number of incidents reported.

The monthly reported calls for both Manhattan and Brooklyn were somewhat similar during the months of June, July, and August.

The incidents reported and the calls for service data are partially correlated by Manhattan and Brooklyn boroughs.

The average response time for the authorities takes for about 40 minutes which is consistent for all five boroughs.

The majority of the calls that were reported for all five boroughs were NON-CIP related incidents, which means that the incidents were reported after it happened.

The incidents that were reported occurred during the daytime making Brooklyn the highest number of incidents reported.

Thank you!